

UID EXP – Student Performance

AIM:

The aim is to create data visualizations such as pie charts and bar graphs to analyze student performance using JavaScript.

PROCEDURE:

Step 1: Create HTML File (index.html)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Performance Visualization</title>

<style>
body {
    font-family: Arial;
    text-align: center;
    margin: 50px;
}
canvas {
    margin: 20px auto;
}
</style>
</head>

<body>

<h1>Student Performance System</h1>

<canvas id="pieChart" width="400" height="400"></canvas>
<canvas id="barChart" width="400" height="400"></canvas>

<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="script.js"></script>
```

```
</body>
</html>
```

Step 2: Create JavaScript File (script.js)

```
// Student performance data
const studentData = {
  labels: ['Math', 'Science', 'English', 'History', 'Computer'],
  datasets: [{
    label: 'Marks',
    data: [85, 90, 75, 80, 95],
    backgroundColor: [
      '#FF6384',
      '#36A2EB',
      '#FFCE56',
      '#4BC0C0',
      '#9966FF'
    ]
  }]
};

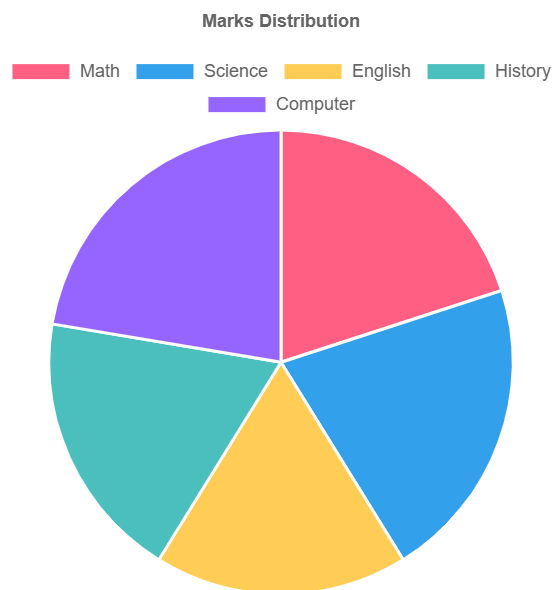
// PIE CHART
const ctxPie = document.getElementById('pieChart').getContext('2d');
new Chart(ctxPie, {
  type: 'pie',
  data: studentData,
  options: {
    responsive: true,
    plugins: {
      title: {
        display: true,
        text: 'Marks Distribution'
      }
    }
  }
});

// BAR CHART
const ctxBar = document.getElementById('barChart').getContext('2d');
new Chart(ctxBar, {
  type: 'bar',
  data: studentData,
  options: {
    responsive: true,
```

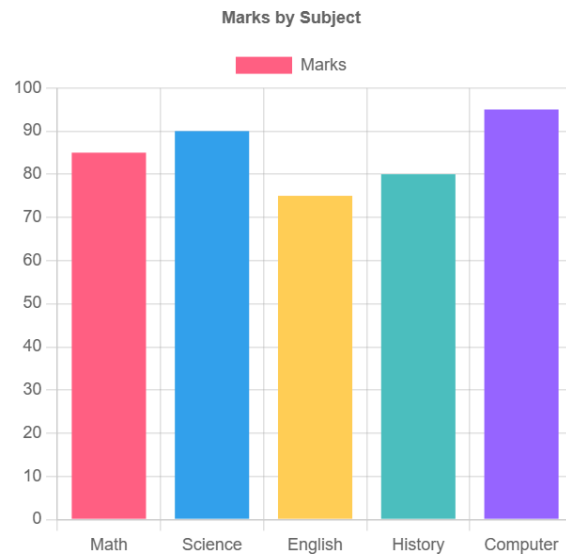
```
plugins: {
  title: {
    display: true,
    text: 'Marks by Subject'
  }
},
scales:
  { y: {
    beginAtZero: true
  }
}
}
});
```

OUTPUT:

Pie chart showing distribution of marks across subjects.



Bar chart comparing marks for each subject.



RESULT:

Thus, the student performance visualization system was successfully created using JavaScript and Chart.js.